AI-Powered Wound Care Assistant

A Computer Vision and Multi-Agent Framework for Wound Monitoring, Evaluated through Agent-Based Simulation

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Semester Project Report

Abstract

Chronic and post-operative wounds need regular monitoring, yet many patients cannot visit a clinician often enough to catch deterioration early. This paper presents an AI-powered wound care assistant that combines a deep-learning image classifier with a multi-agent decision pipeline. A **Vision Agent** uses a MobileNetV2 transfer-learning model to label an image and report class probabilities. A **Comparison Agent** stores each patient's history and decides whether the condition is a baseline, stable, improving, worsening or changed. An **Advisory Agent** combines both outputs into an urgency level (LOW, MEDIUM or HIGH) with plain-language advice, and an orchestrator runs the three agents in sequence. To study how the system behaves over time, we extend it into an agent-based model in which a **Patient Agent** simulates daily wound healing, with random healing rates and infection events, and a **Doctor Agent** receives alerts. In a 30-day simulation of 300 patients, 78 patients became critical. With the agents, the doctor was alerted for all 78 of them, on average 3.6 days *before* the wound crossed the critical threshold. Without the agents, which relies on patient self-reporting and routine visits, 93.6% were alerted, on average 2.9 days *after* the threshold. The agents raised 18 alerts for patients who never became critical. These results come from a simulation with assumed parameters and show the potential benefit of continuous automated monitoring, not clinical performance.

Keywords: wound monitoring, multi-agent systems, agentic AI, MobileNetV2, transfer learning, agent-based simulation, telemedicine

1. Introduction

Wounds that heal slowly or become infected are a significant burden on patients and healthcare systems. Early signs of deterioration, such as spreading redness, swelling or changes in tissue appearance, are often visible days before a patient decides to seek care. In practice, patients tend to notice a problem late, and follow-up appointments are infrequent. Delays of even a few days can turn a manageable infection into a serious one.

Recent advances in computer vision make it possible to analyse a photograph taken with an ordinary smartphone. However, a classifier on its own answers only one question: *what does this image look like right now?* A useful care assistant must also answer *is the wound getting better or worse than before?* and *what should the patient do?* These are separate tasks that need memory, comparison and decision rules, which fits naturally with an agent-based design in which each agent has a single responsibility and passes its output to the next.

This project builds such a system and then evaluates it over time. The contributions of this work are:

- A perception agent based on MobileNetV2 transfer learning that returns a label, a confidence score and the scores of all classes.
- A memory agent that keeps a patient's history and classifies the trend (baseline, stable, improving, worsening or changed) using explicit rules.
- A decision agent that converts the perception and trend outputs into an urgency level and advice, and an orchestrator that coordinates the three agents.
- An agent-based simulation with a Patient Agent (random healing and infection) and a Doctor Agent, used to compare doctor-alert timing with and without the agents over 30 days.

The remainder of this paper is organised as follows. Section 2 reviews the background. Section 3 describes the system architecture and the three agents. Section 4 presents the agent-based simulation. Section 5 reports the experimental setup and results. Section 6 discusses limitations, and Section 7 concludes with future work.

2. Background and Related Work

Transfer learning for medical images. Medical image datasets are usually small compared with general image collections, so models pretrained on ImageNet are commonly reused as feature extractors. MobileNetV2 (Sandler et al., 2018) uses inverted residual blocks with linear bottlenecks, giving good accuracy at a low computational cost, which suits deployment on phones or small servers.

Dataset. Real wound datasets with reliable labels are rare and small. We therefore use HAM10000 (Tschandl et al., 2018), a public collection of dermatoscopic skin lesion images covering seven diagnostic categories, as a proxy dataset. The classification pipeline and agent logic are independent of the dataset and could be retrained on wound images when available.

Intelligent agents. An agent perceives its environment, keeps some internal state and acts to achieve a goal (Wooldridge and Jennings, 1995; Russell and Norvig, 2021). Dividing a clinical workflow into perception, memory and decision agents keeps each component simple to test and explain.

Agent-based modelling. Agent-based models represent a system as autonomous agents that interact over time, so that overall behaviour emerges from individual rules and random events (Bonabeau, 2002). This is well suited to studying how a monitoring system performs across many patients whose wounds heal at different speeds.

3. System Architecture

The system is a pipeline of three agents controlled by an orchestrator (Figure 1). Each agent has one goal, one set of inputs and one type of output.

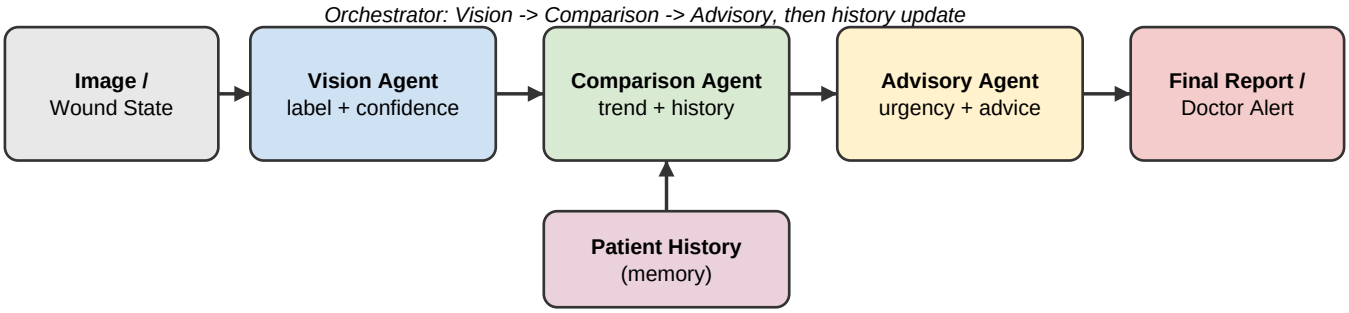


Figure 1. Multi-agent pipeline: Vision, Comparison and Advisory agents coordinated by the orchestrator, with patient history kept as memory.

3.1 Vision Agent (Perception)

The Vision Agent resizes the input image to 224 x 224, applies the MobileNetV2 preprocessing and passes it to the trained network. The model uses a frozen MobileNetV2 base pretrained on ImageNet, followed by global average pooling, a dense layer of 128 units with ReLU, dropout of 0.3 and a softmax output with one unit per class. The agent selects the class with the highest probability and returns the label, its confidence and the scores of all classes. Its state is the set of trained model weights.

Setting	Value
Dataset	HAM10000 (7 classes), 80/20 stratified train/validation split
Base network	MobileNetV2, ImageNet weights, frozen
Classification head	GlobalAveragePooling, Dense(128, ReLU), Dropout(0.3), Dense(7, softmax)
Input size / batch size	224 x 224 / 32
Optimizer / loss	Adam (learning rate 0.001) / categorical cross-entropy
Epochs	10
Augmentation	Rotation up to 20 degrees, horizontal flip, zoom up to 10%

Table 1. Configuration of the computer vision model.

3.2 Comparison Agent (Memory and Reasoning)

The Comparison Agent receives the new Vision Agent result and the patient's stored history. It compares the new result with the most recent previous result using the following rules, applied in order:

Condition	Trend status
No previous scan exists	baseline
Same label and confidence rose by more than 5 percentage points	worsening
Same label and confidence fell by more than 5 percentage points	improving
Label differs from the previous scan	changed
Otherwise	stable

Table 2. Decision rules of the Comparison Agent.

3.3 Advisory Agent (Decision)

The Advisory Agent takes the Vision Agent and Comparison Agent outputs and returns an urgency level with advice. A prediction of a high-risk label with confidence above 60% gives urgency **HIGH** and advice to contact a doctor immediately. A worsening trend gives **MEDIUM**. An improving trend gives **LOW** with advice to continue care. In the simulation, a label change to a more severe class is treated as worsening and a change to a milder class as improving, because the simulated classes have a natural severity order.

3.4 Orchestrator

The orchestrator runs the agents in the sequence Vision, Comparison, Advisory, returns a final report containing all three outputs, and then appends the new Vision Agent result to the patient's history so that the next scan is compared against it.

4. Agent-Based Simulation

To study how the pipeline performs over time, the system is placed in a 30-day simulation. Each simulated day, every patient's wound state changes, the patient is monitored by the agents, and alerts reach the doctor.

4.1 Patient Agent

Each Patient Agent holds a wound severity value between 0 (healed) and 1 (most severe), starting between 0.20 and 0.40, and an infection flag. Every patient receives random individual parameters: a healing rate, a daily infection probability, an infection growth rate, a daily probability that the immune system clears the infection, a symptom-awareness threshold and a care-seeking delay of one to five days. Each day, an infection may begin or clear at random. When infected, severity increases by a random amount and healing is reduced. When not infected, severity decreases in proportion to the healing rate. A wound is considered *critical* on the first day its severity reaches 0.65.

4.2 Monitored and Unmonitored Scenarios

In the **with-agents** scenario, the patient is scanned daily. A simulated Vision Agent turns the true severity into class probabilities over four classes (Healing, Mild Inflammation, Infected, Severe Infection) with added noise, imitating an imperfect classifier. The Comparison and Advisory agents then run as described in Section 3, and the Doctor Agent is alerted on the first day the urgency is HIGH. In the **without-agents** scenario, the doctor learns of a problem only when the patient notices symptoms (severity above their awareness threshold) and reports after their care-seeking delay, or at routine visits on days 14 and 28 if the wound is then critical. The first of these events is the alert day. Both scenarios observe the same patient trajectory, so the comparison is paired.

4.3 Doctor Agent and Metrics

The Doctor Agent records the first alert day and its source for each patient. The main metric is the *alert delay*, defined as the alert day minus the critical day, so a negative value means the doctor was warned before the wound became critical. We also report the percentage of critical patients alerted within 30 days and the number of alerts raised for patients who never became critical (false alerts).

5. Experiments and Results

5.1 Setup

Parameter	Value
Simulation length	30 days
Number of patients	300 (random seed 42)
Critical severity threshold	0.65
Routine clinic visits (without agents)	Days 14 and 28
Doctor alert level (with agents)	Urgency HIGH
Classes in the simulated Vision Agent	Healing, Mild Inflammation, Infected, Severe Infection

Table 3. Simulation parameters.

5.2 Single-Patient Example

Figure 2 follows one patient through the 30 days. The wound severity rises after an infection starts. The Advisory Agent switches from LOW to HIGH urgency on day 6, three days before the wound becomes critical on day 9, and the doctor is alerted that day. Without the agents, the doctor would only learn of the problem on day 13, after the patient notices symptoms and reports them.

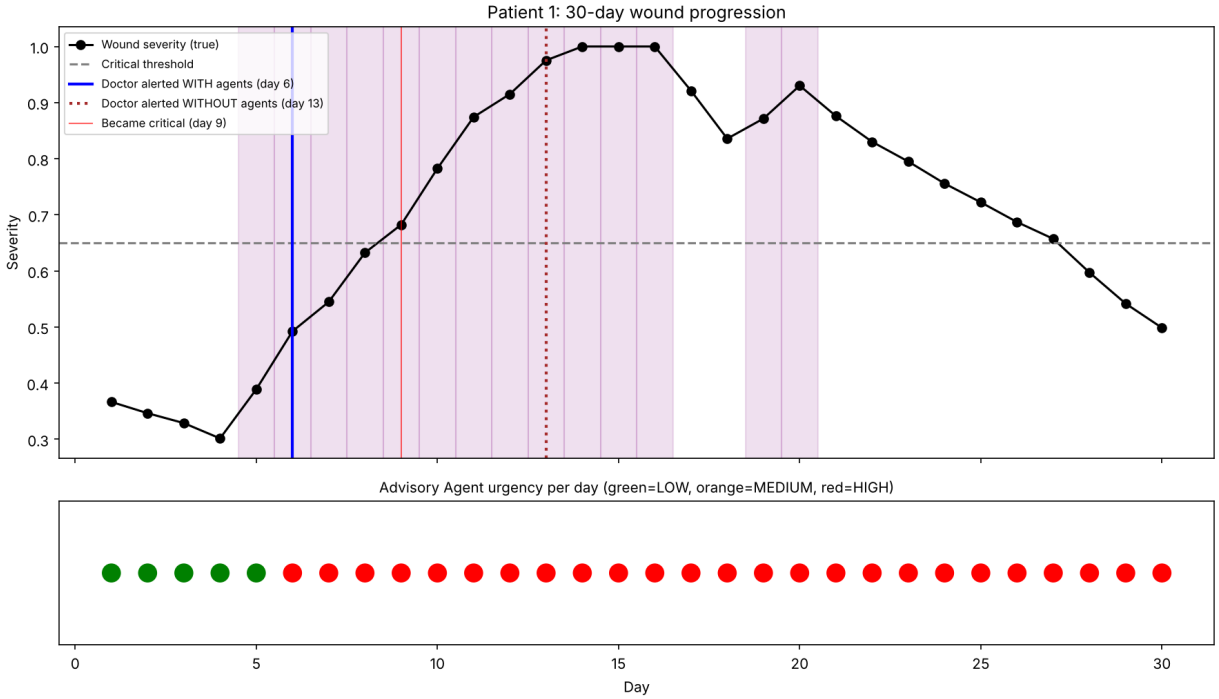


Figure 2. Thirty-day course of one simulated patient: severity, infection periods, daily urgency from the Advisory Agent, and the doctor-alert day with and without agents.

5.3 Population Results

Of the 300 simulated patients, 78 became critical within 30 days. Table 4 and Figure 3 compare the two scenarios for these patients.

Metric	With agents	Without agents
Critical patients alerted within 30 days	100.0%	93.6%
Mean alert delay (days from critical day)	-3.58	+2.95
Median alert delay (days from critical day)	-3.00	+3.00
Alerts for patients who never became critical	18	0

Table 4. Doctor-alert timing for the 78 patients who became critical (300 simulated patients, 30 days).

With the agents, every critical patient was alerted, on average 3.6 days before the wound reached the critical threshold. Without the agents, alerts arrived on average 2.9 days after the threshold, and about 6% of critical patients were never alerted within the 30 days. The cumulative curve in Figure 3 shows that the agents also alert a larger share of patients at every day of the simulation. The cost of earlier alerts is false alarms: 18 of the 222 patients who never became critical received a HIGH alert.

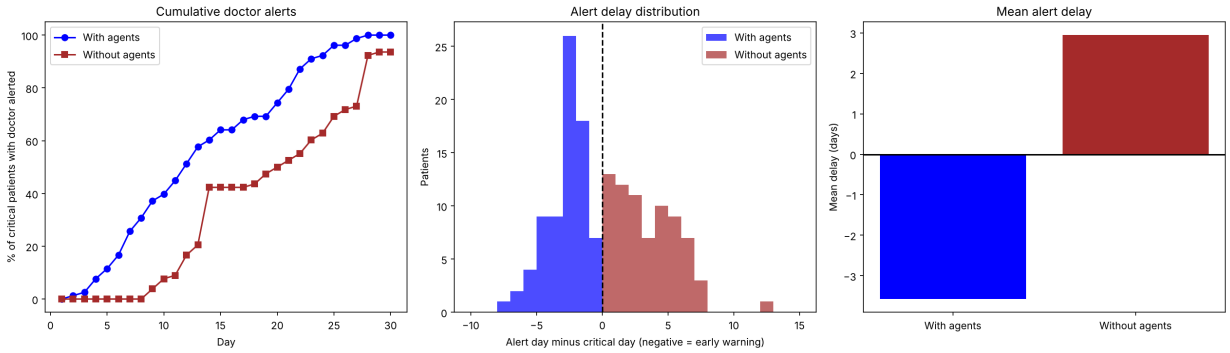


Figure 3. Comparison of doctor alerts with and without agents: cumulative percentage of critical patients alerted (left), distribution of alert delay (centre) and mean alert delay (right).

6. Discussion and Limitations

The results show that daily automated monitoring can shift the first doctor alert from after deterioration to before it, in a setting where patients otherwise rely on noticing symptoms themselves. The agents are not free of cost, as the earlier alerts come with a small false-alarm rate. Alert thresholds such as the 60% confidence limit and the choice of which urgency levels notify the doctor are therefore design decisions that trade early warning against unnecessary alerts.

The following limitations should be kept in mind:

- The simulation parameters (healing rates, infection probabilities, thresholds and patient behaviour) are assumptions, not values estimated from clinical data. The numbers in Section 5 illustrate the behaviour of the system and are not clinical evidence.
- In the simulation, the Vision Agent is a noisy function of the true severity and not the trained CNN, because no daily wound images exist for simulated patients.
- HAM10000 contains skin lesions, not wounds, and is used as a proxy. Its classes (for example melanoma) do not describe wound healing, so the CNN would need to be retrained on a wound dataset for real deployment.
- The Comparison Agent's rules use changes in model confidence as a proxy for progression. Confidence is not a direct measure of wound severity, and a calibrated severity score would be more reliable.
- The system is a decision-support prototype and does not replace medical diagnosis.

7. Conclusion and Future Work

We presented a wound care assistant built from three single-purpose agents (Vision, Comparison and Advisory) coordinated by an orchestrator, and extended it with Patient and Doctor agents to simulate 30 days of wound progression. In the simulation, continuous agent-based monitoring alerted the doctor on average 3.6 days before a wound became critical, whereas patient self-reporting and routine visits alerted on average 2.9 days after, at the cost of a small number of false alerts.

Future work includes: training the vision model on a real wound dataset with severity or infection labels; replacing the simulated Vision Agent with the trained model in the simulation; modelling the effect of treatment after an alert; replacing the rule-based Advisory Agent with a language model for more natural advice; adding a reminder agent for follow-up checks; storing history in a database; and deploying a web interface for patients.

References

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